

Titration report



Method

Method ID	CDC_GranTED	Release state	Released
Name	Titration with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/3/2026 2:54:52 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013151 mL
		Concentration R2	0.10026 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.015980 mL
		Concentration R2	0.10032 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013259 mL ⁽¹⁶⁾
		Concentration R2	0.10027 mol/L ⁽¹⁶⁾
4	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013073 mL
		Concentration R2	0.10026 mol/L
5	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013240 mL
		Concentration R2	0.10026 mol/L
6	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013567 mL
		Concentration R2	0.10027 mol/L

Statistics

R1

Mean	5.013712 mL ⁽¹⁶⁾	Minimum	5.013073 mL
Standard deviation	0.001124 mL ⁽¹⁶⁾	Rel. standard deviation	0.022 % ⁽¹⁶⁾
Total samples	6	Samples excluded	0

R2

Mean	0.10027 mol/L ⁽¹⁶⁾	Minimum	0.10026 mol/L
Standard deviation	0.00002 mol/L ⁽¹⁶⁾	Rel. standard deviation	0.023 % ⁽¹⁶⁾
Total samples	6	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
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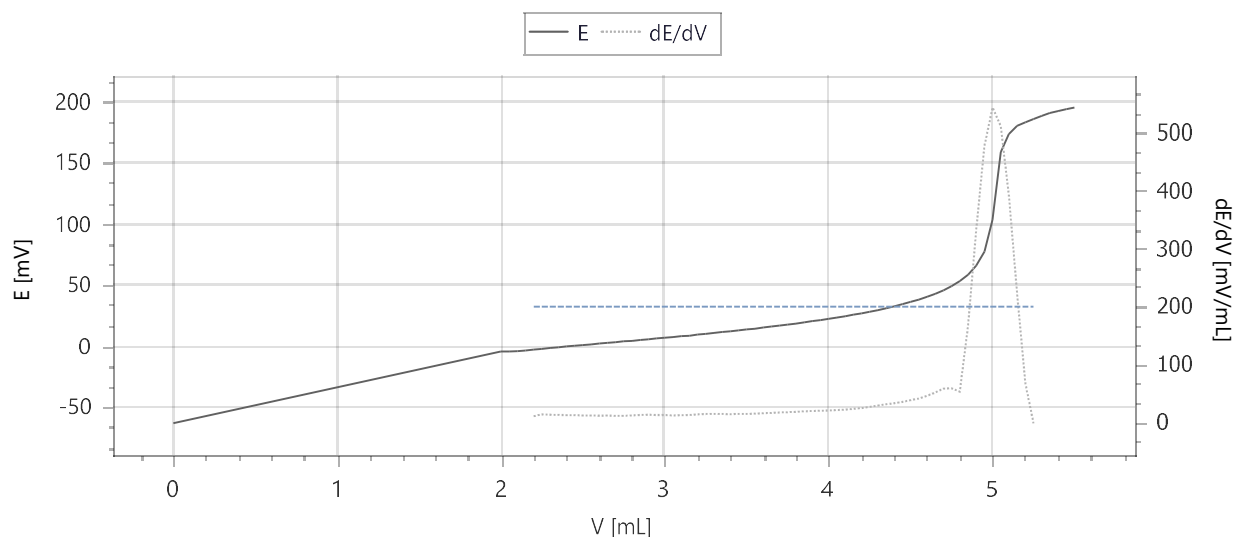
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LabX

0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.013151 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.10026 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.015980 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.10032 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption ⁽¹⁶⁾	R1 ⁽¹⁶⁾	5.013259 mL ⁽¹⁶⁾	OK ⁽¹⁶⁾
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration ⁽¹⁶⁾	R2 ⁽¹⁶⁾	0.10027 mol/L ⁽¹⁶⁾	OK ⁽¹⁶⁾
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.013073 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.10026 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.013240 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.10026 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.013567 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.10027 mol/L	OK

Titration (EQP) [1] Scope 1/1, Sample 1/6

E - V & dE/dV - V Curve Scope 1/1, Sample 1/6



Measured Values Scope 1/1, Sample 1/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-63.0		25.0			
10	2.0000	-4.3	2.0000	25.0	58.7		
43	2.0500	-4.4	0.0500	25.0	-0.1		
46	2.1000	-3.9	0.0500	25.0	0.5		

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50	2.1500	-3.4	0.0500	25.0	0.5	
53	2.2000	-2.8	0.0500	25.0	0.6	10.62
57	2.2500	-2.1	0.0500	25.0	0.7	13.66
60	2.3000	-1.5	0.0500	25.0	0.6	13.00
65	2.3500	-0.9	0.0500	25.0	0.6	12.74
68	2.4000	-0.1	0.0500	25.0	0.8	12.32
71	2.4500	0.4	0.0500	25.0	0.5	12.29
74	2.5000	0.9	0.0500	25.0	0.5	11.78
78	2.5500	1.6	0.0500	25.0	0.7	11.84
81	2.6000	2.2	0.0500	25.0	0.6	11.35
84	2.6500	2.8	0.0500	25.0	0.6	11.59
88	2.7000	3.2	0.0500	25.0	0.4	11.09
91	2.7500	4.0	0.0500	25.0	0.8	11.16
94	2.8000	4.3	0.0500	25.0	0.3	11.89
97	2.8500	5.1	0.0500	25.0	0.8	12.53
100	2.9000	5.6	0.0500	25.0	0.5	13.16
104	2.9500	6.4	0.0500	25.0	0.8	12.47
107	3.0000	7.0	0.0500	25.0	0.6	12.52
110	3.0500	7.6	0.0500	25.0	0.6	11.76
114	3.1000	8.2	0.0500	25.0	0.6	12.16
117	3.1500	8.6	0.0500	25.0	0.4	12.57
120	3.2000	9.5	0.0500	25.0	0.9	13.63
124	3.2500	10.1	0.0500	25.0	0.6	14.31
127	3.3000	10.9	0.0500	25.0	0.8	14.65
131	3.3500	11.6	0.0500	25.0	0.7	14.31
134	3.4000	12.3	0.0500	25.0	0.7	13.97
138	3.4500	13.0	0.0500	25.0	0.7	14.41
142	3.5000	13.7	0.0500	25.0	0.7	14.45
146	3.5500	14.4	0.0500	25.0	0.7	15.00
149	3.6000	15.3	0.0500	25.0	0.9	15.60
153	3.6500	16.0	0.0500	25.0	0.7	16.17
156	3.7000	16.8	0.0500	25.0	0.8	16.92
159	3.7500	17.7	0.0500	25.0	0.9	17.40
163	3.8000	18.6	0.0500	25.0	0.9	18.08
166	3.8500	19.5	0.0500	25.0	0.9	18.92
170	3.9000	20.5	0.0500	25.0	1.0	19.40
174	3.9500	21.4	0.0500	25.0	0.9	20.00
178	4.0000	22.5	0.0500	25.0	1.1	20.37
182	4.0500	23.5	0.0500	25.0	1.0	21.07
186	4.1000	24.6	0.0500	25.0	1.1	21.75
190	4.1500	25.7	0.0500	25.0	1.1	23.16
194	4.2000	26.8	0.0500	25.0	1.1	24.54
198	4.2500	28.2	0.0500	25.0	1.4	26.56
201	4.3000	29.5	0.0500	25.0	1.3	29.00
206	4.3500	31.1	0.0500	25.0	1.6	31.06
210	4.4000	32.6	0.0500	25.0	1.5	33.03
213	4.4500	34.4	0.0500	25.0	1.8	35.22
218	4.5000	36.3	0.0500	25.0	1.9	38.15
222	4.5500	38.1	0.0500	25.0	1.8	41.11
227	4.6000	40.4	0.0500	25.0	2.3	45.66
231	4.6500	42.9	0.0500	25.0	2.5	51.48
236	4.7000	45.7	0.0500	25.0	2.8	58.17

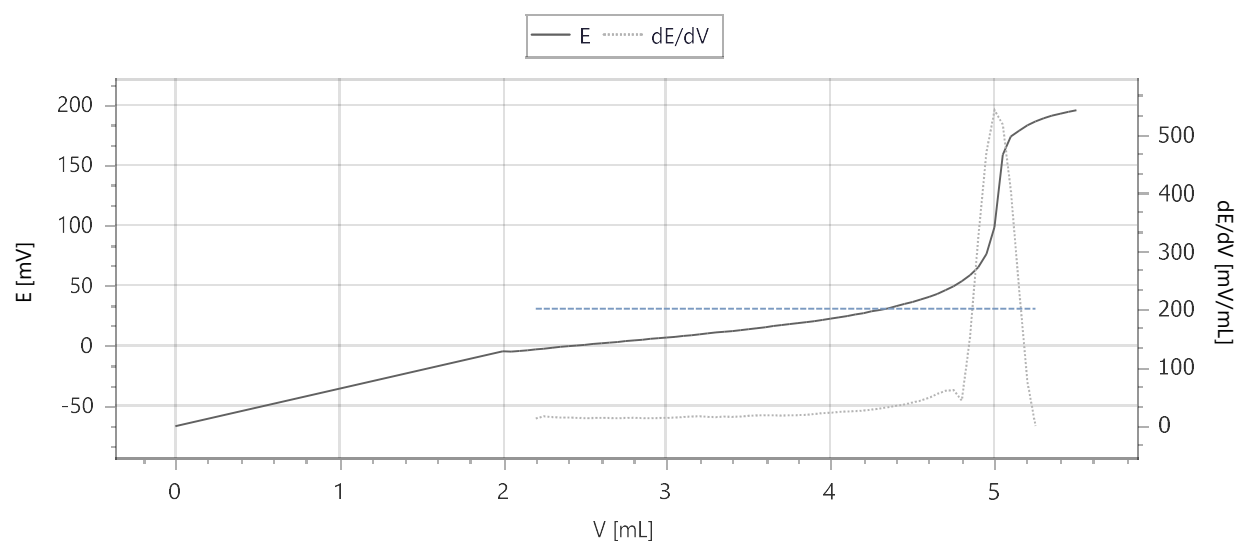
Titration report

LabX

240	4.7500	49.0	0.0500	25.0	3.3	58.72	
245	4.8000	53.3	0.0500	25.0	4.3	52.91	
251	4.8500	58.6	0.0500	25.0	5.3	166.06	
259	4.9000	65.8	0.0500	25.0	7.2	331.89	
269	4.9500	77.4	0.0500	25.0	11.6	477.11	
275	5.0000	103.3	0.0500	25.0	25.9	544.19	
	5.013151	117.9				544.37	EQP
281	5.0500	158.7	0.0500	25.0	55.4	511.87	
286	5.1000	173.5	0.0500	25.0	14.8	390.82	
290	5.1500	180.3	0.0500	25.0	6.8	221.63	
298	5.2000	183.0	0.0500	25.0	2.7	67.36	
304	5.2500	185.8	0.0500	25.0	2.8	-1.82	
308	5.3000	188.4	0.0500	25.0	2.6		
311	5.3500	190.6	0.0500	25.0	2.2		
315	5.4000	192.3	0.0500	25.0	1.7		
318	5.4500	193.8	0.0500	25.0	1.5		
322	5.5000	195.2	0.0500	25.0	1.4		

Titration (EQP) [1] Scope 1/1, Sample 2/6

E - V & dE/dV - V Curve Scope 1/1, Sample 2/6



Measured Values Scope 1/1, Sample 2/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-66.6		25.0			
10	2.0000	-4.6	2.0000	25.0	62.0		
43	2.0500	-4.9	0.0500	25.0	-0.3		
46	2.1000	-4.5	0.0500	25.0	0.4		
50	2.1500	-3.8	0.0500	25.0	0.7		
53	2.2000	-3.1	0.0500	25.0	0.7	11.04	
56	2.2500	-2.5	0.0500	25.0	0.6	14.86	
59	2.3000	-1.8	0.0500	25.0	0.7	13.45	
63	2.3500	-1.0	0.0500	25.0	0.8	12.71	
66	2.4000	-0.4	0.0500	25.0	0.6	12.70	

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LabX

69	2.4500	0.0	0.0500	25.0	0.4	12.18
72	2.5000	0.7	0.0500	25.0	0.7	11.58
76	2.5500	1.4	0.0500	25.0	0.7	11.91
79	2.6000	2.0	0.0500	25.0	0.6	12.07
82	2.6500	2.4	0.0500	25.0	0.4	11.91
85	2.7000	3.1	0.0500	25.0	0.7	11.40
88	2.7500	3.8	0.0500	25.0	0.7	11.99
91	2.8000	4.2	0.0500	25.0	0.4	12.42
94	2.8500	4.9	0.0500	25.0	0.7	11.68
98	2.9000	5.5	0.0500	25.0	0.6	11.49
101	2.9500	6.2	0.0500	25.0	0.7	11.88
104	3.0000	6.6	0.0500	25.0	0.4	12.13
107	3.0500	7.1	0.0500	25.0	0.5	12.63
110	3.1000	8.1	0.0500	25.0	1.0	13.63
113	3.1500	8.6	0.0500	25.0	0.5	14.48
116	3.2000	9.3	0.0500	25.0	0.7	14.80
120	3.2500	10.2	0.0500	25.0	0.9	13.89
122	3.3000	10.8	0.0500	25.0	0.6	13.42
126	3.3500	11.5	0.0500	25.0	0.7	14.25
129	3.4000	12.1	0.0500	25.0	0.6	13.93
132	3.4500	12.8	0.0500	25.0	0.7	14.35
135	3.5000	13.7	0.0500	25.0	0.9	15.67
138	3.5500	14.5	0.0500	25.0	0.8	16.29
141	3.6000	15.1	0.0500	25.0	0.6	16.52
144	3.6500	16.1	0.0500	25.0	1.0	16.46
147	3.7000	17.0	0.0500	25.0	0.9	16.13
150	3.7500	17.7	0.0500	25.0	0.7	16.46
153	3.8000	18.5	0.0500	25.0	0.8	16.81
156	3.8500	19.4	0.0500	25.0	0.9	17.33
159	3.9000	20.2	0.0500	25.0	0.8	18.77
162	3.9500	21.3	0.0500	25.0	1.1	20.35
165	4.0000	22.3	0.0500	25.0	1.0	20.85
169	4.0500	23.4	0.0500	25.0	1.1	22.34
173	4.1000	24.5	0.0500	25.0	1.1	23.22
176	4.1500	25.7	0.0500	25.0	1.2	23.60
180	4.2000	26.8	0.0500	25.0	1.1	24.68
182	4.2500	28.4	0.0500	25.0	1.6	26.37
186	4.3000	29.4	0.0500	25.0	1.0	28.19
189	4.3500	30.9	0.0500	25.0	1.5	30.49
193	4.4000	32.6	0.0500	25.0	1.7	32.86
197	4.4500	34.4	0.0500	25.0	1.8	35.01
200	4.5000	36.1	0.0500	25.0	1.7	38.73
204	4.5500	38.2	0.0500	25.0	2.1	41.67
209	4.6000	40.3	0.0500	25.0	2.1	46.53
212	4.6500	42.8	0.0500	25.0	2.5	53.18
215	4.7000	45.9	0.0500	25.0	3.1	58.61
218	4.7500	49.1	0.0500	25.0	3.2	59.91
224	4.8000	53.3	0.0500	25.0	4.2	42.03
231	4.8500	58.3	0.0500	25.0	5.0	150.09
236	4.9000	64.7	0.0500	25.0	6.4	320.26
244	4.9500	75.8	0.0500	25.0	11.1	468.38
247	5.0000	98.1	0.0500	25.0	22.3	541.09

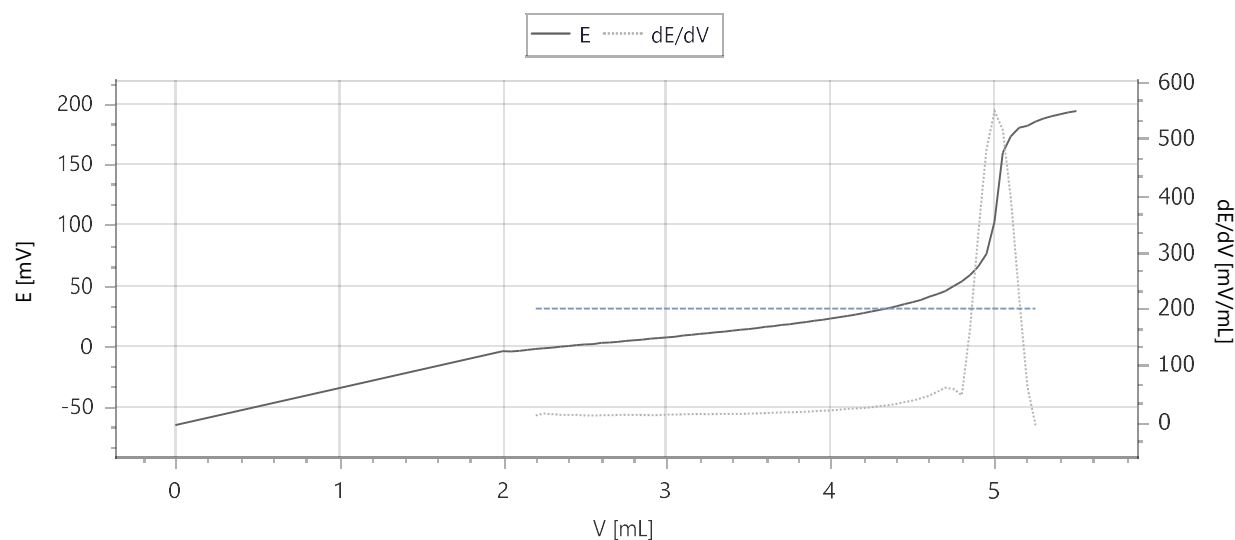
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LabX

	5.015980	117.2				541.64	EQP
254	5.0500	157.9	0.0500	25.0	59.8	516.49	
258	5.1000	172.9	0.0500	25.0	15.0	401.18	
269	5.1500	177.8	0.0500	25.0	4.9	232.74	
273	5.2000	182.2	0.0500	25.0	4.4	74.99	
277	5.2500	185.5	0.0500	25.0	3.3	-2.06	
280	5.3000	188.0	0.0500	25.0	2.5		
283	5.3500	190.1	0.0500	25.0	2.1		
286	5.4000	191.9	0.0500	25.0	1.8		
289	5.4500	193.4	0.0500	25.0	1.5		
292	5.5000	194.7	0.0500	25.0	1.3		

Titration (EQP) [1] Scope 1/1, Sample 3/6

E - V & dE/dV - V Curve Scope 1/1, Sample 3/6



Measured Values Scope 1/1, Sample 3/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-65.9		25.0			
9	2.0000	-4.6	2.0000	25.0	61.3		
43	2.0500	-4.8	0.0500	25.0	-0.2		
47	2.1000	-4.4	0.0500	25.0	0.4		
50	2.1500	-3.6	0.0500	25.0	0.8		
53	2.2000	-2.8	0.0500	25.0	0.8	11.52	
56	2.2500	-2.3	0.0500	25.0	0.5	14.74	
59	2.3000	-1.7	0.0500	25.0	0.6	13.71	
62	2.3500	-0.8	0.0500	25.0	0.9	12.55	
66	2.4000	-0.3	0.0500	25.0	0.5	12.59	
69	2.4500	0.4	0.0500	25.0	0.7	12.34	
72	2.5000	0.9	0.0500	25.0	0.5	11.43	
76	2.5500	1.4	0.0500	25.0	0.5	11.17	
79	2.6000	2.2	0.0500	25.0	0.8	11.71	
83	2.6500	2.6	0.0500	25.0	0.4	11.73	
86	2.7000	3.2	0.0500	25.0	0.6	12.24	

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89	2.7500	3.9	0.0500	25.0	0.7	12.66	
92	2.8000	4.5	0.0500	25.0	0.6	12.46	
95	2.8500	5.1	0.0500	25.0	0.6	12.19	
98	2.9000	5.8	0.0500	25.0	0.7	12.03	
102	2.9500	6.4	0.0500	25.0	0.6	12.04	
105	3.0000	6.8	0.0500	25.0	0.4	12.80	
108	3.0500	7.5	0.0500	25.0	0.7	13.04	
111	3.1000	8.4	0.0500	25.0	0.9	13.38	
114	3.1500	8.9	0.0500	25.0	0.5	13.70	
117	3.2000	9.7	0.0500	25.0	0.8	13.96	
120	3.2500	10.2	0.0500	25.0	0.5	13.68	
123	3.3000	11.0	0.0500	25.0	0.8	13.89	
126	3.3500	11.7	0.0500	25.0	0.7	13.99	
130	3.4000	12.5	0.0500	25.0	0.8	14.32	
134	3.4500	13.1	0.0500	25.0	0.6	14.27	
137	3.5000	13.8	0.0500	25.0	0.7	14.59	
140	3.5500	14.5	0.0500	25.0	0.7	15.01	
143	3.6000	15.5	0.0500	25.0	1.0	15.56	
146	3.6500	16.0	0.0500	25.0	0.5	16.32	
149	3.7000	17.1	0.0500	25.0	1.1	16.73	
152	3.7500	17.7	0.0500	25.0	0.6	17.07	
156	3.8000	18.7	0.0500	25.0	1.0	17.38	
160	3.8500	19.6	0.0500	25.0	0.9	18.10	
163	3.9000	20.5	0.0500	25.0	0.9	18.66	
166	3.9500	21.4	0.0500	25.0	0.9	19.92	
169	4.0000	22.4	0.0500	25.0	1.0	20.13	
172	4.0500	23.5	0.0500	25.0	1.1	21.62	
175	4.1000	24.7	0.0500	25.0	1.2	23.02	
178	4.1500	25.7	0.0500	25.0	1.0	23.85	
182	4.2000	26.9	0.0500	25.0	1.2	24.48	
186	4.2500	28.4	0.0500	25.0	1.5	25.76	
190	4.3000	29.6	0.0500	25.0	1.2	27.95	
193	4.3500	30.9	0.0500	25.0	1.3	29.28	
198	4.4000	32.5	0.0500	25.0	1.6	31.76	
203	4.4500	34.3	0.0500	25.0	1.8	34.92	
208	4.5000	36.1	0.0500	25.0	1.8	37.73	
211	4.5500	37.8	0.0500	25.0	1.7	41.53	
214	4.6000	40.5	0.0500	25.0	2.7	46.28	
217	4.6500	42.7	0.0500	25.0	2.2	53.07	
221	4.7000	45.4	0.0500	25.0	2.7	60.34	
226	4.7500	49.3	0.0500	25.0	3.9	57.98	
232	4.8000	53.3	0.0500	25.0	4.0	47.32	
240	4.8500	58.6	0.0500	25.0	5.3	163.15	
245	4.9000	65.4	0.0500	25.0	6.8	330.45	
253	4.9500	76.1	0.0500	25.0	10.7	479.58	
256	5.0000	102.5	0.0500	25.0	26.4	548.25	
	5.013259	117.7				548.45	EQP
263	5.0500	159.9	0.0500	25.0	57.4	514.98	
266	5.1000	173.5	0.0500	25.0	13.6	392.66	
270	5.1500	180.6	0.0500	25.0	7.1	220.05	
281	5.2000	182.2	0.0500	25.0	1.6	62.30	
284	5.2500	185.5	0.0500	25.0	3.3	-5.79	

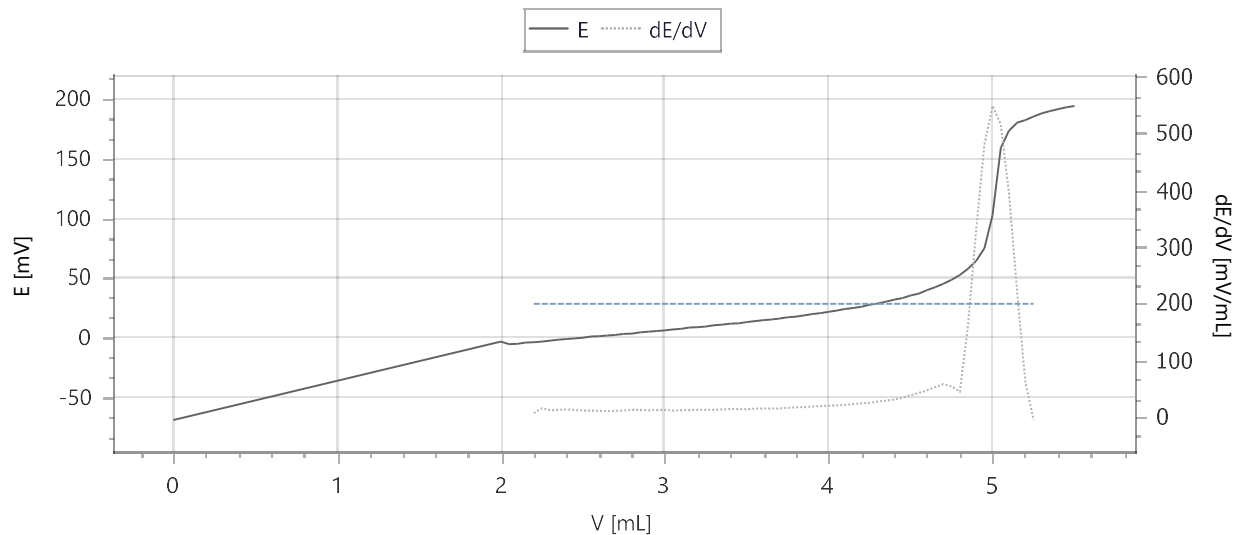
Titration report

LabX

287	5.3000	188.2	0.0500	25.0	2.7
291	5.3500	190.1	0.0500	25.0	1.9
294	5.4000	191.8	0.0500	25.0	1.7
298	5.4500	193.2	0.0500	25.0	1.4
300	5.5000	194.4	0.0500	25.0	1.2

Titration (EQP) [1] Scope 1/1, Sample 4/6

E - V & dE/dV - V Curve Scope 1/1, Sample 4/6



Measured Values Scope 1/1, Sample 4/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-68.6		25.0			
6	2.0000	-3.1	2.0000	25.0	65.5		
39	2.0500	-5.3	0.0500	25.0	-2.2		
42	2.1000	-4.9	0.0500	25.0	0.4		
46	2.1500	-3.9	0.0500	25.0	1.0		
49	2.2000	-3.7	0.0500	25.0	0.2	7.46	
52	2.2500	-3.1	0.0500	25.0	0.6	16.23	
56	2.3000	-2.3	0.0500	25.0	0.8	12.92	
59	2.3500	-1.5	0.0500	25.0	0.8	13.16	
62	2.4000	-1.0	0.0500	25.0	0.5	14.12	
65	2.4500	-0.3	0.0500	25.0	0.7	13.00	
68	2.5000	0.2	0.0500	25.0	0.5	12.06	
71	2.5500	1.1	0.0500	25.0	0.9	12.03	
74	2.6000	1.6	0.0500	25.0	0.5	11.42	
78	2.6500	2.0	0.0500	25.0	0.4	10.94	
81	2.7000	2.6	0.0500	25.0	0.6	11.55	
84	2.7500	3.3	0.0500	25.0	0.7	12.14	
87	2.8000	3.7	0.0500	25.0	0.4	13.23	
90	2.8500	4.6	0.0500	25.0	0.9	13.18	
93	2.9000	5.3	0.0500	25.0	0.7	12.44	
96	2.9500	5.8	0.0500	25.0	0.5	12.93	
99	3.0000	6.4	0.0500	25.0	0.6	13.17	

Titration report

LabX

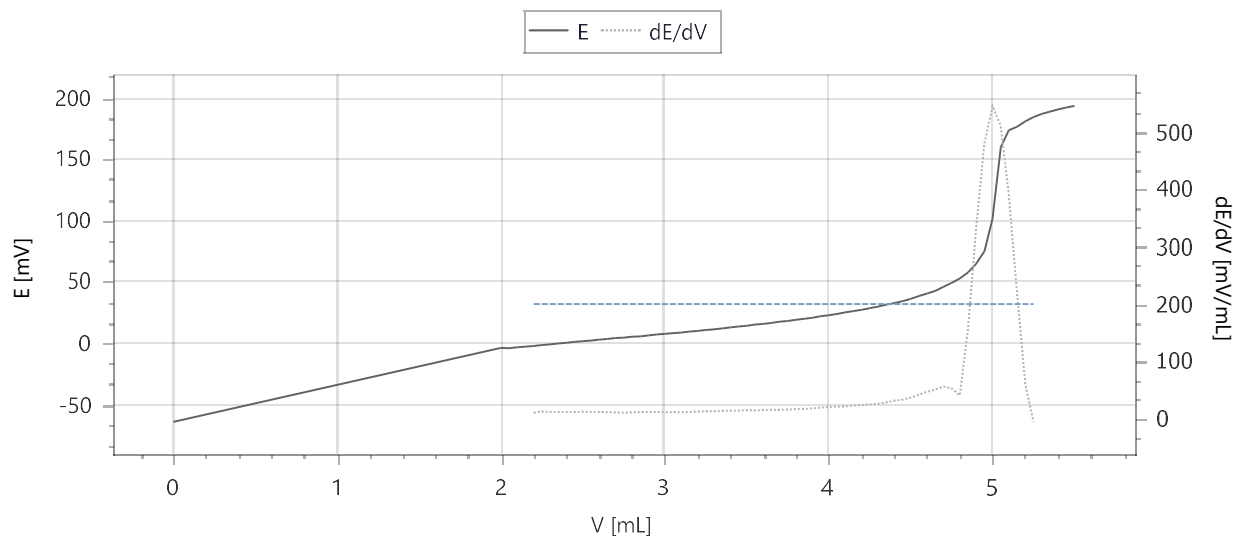
102	3.0500	7.0	0.0500	25.0	0.6	11.96	EQP
106	3.1000	7.7	0.0500	25.0	0.7	12.53	
109	3.1500	8.6	0.0500	25.0	0.9	13.02	
112	3.2000	8.9	0.0500	25.0	0.3	13.46	
115	3.2500	9.4	0.0500	25.0	0.5	13.33	
118	3.3000	10.5	0.0500	25.0	1.1	13.39	
121	3.3500	11.1	0.0500	25.0	0.6	13.94	
124	3.4000	11.8	0.0500	25.0	0.7	14.86	
128	3.4500	12.3	0.0500	25.0	0.5	14.62	
130	3.5000	13.2	0.0500	25.0	0.9	14.21	
134	3.5500	14.0	0.0500	25.0	0.8	15.76	
137	3.6000	14.8	0.0500	25.0	0.8	15.68	
140	3.6500	15.5	0.0500	25.0	0.7	15.66	
143	3.7000	16.2	0.0500	25.0	0.7	15.81	
147	3.7500	17.3	0.0500	25.0	1.1	16.89	
150	3.8000	17.8	0.0500	25.0	0.5	17.65	
154	3.8500	18.9	0.0500	25.0	1.1	18.12	
157	3.9000	19.9	0.0500	25.0	1.0	19.12	
160	3.9500	20.8	0.0500	25.0	0.9	19.85	
163	4.0000	21.7	0.0500	25.0	0.9	20.70	
166	4.0500	22.8	0.0500	25.0	1.1	21.10	
169	4.1000	24.1	0.0500	25.0	1.3	22.14	
173	4.1500	25.0	0.0500	25.0	0.9	23.45	
176	4.2000	26.2	0.0500	25.0	1.2	24.85	
179	4.2500	27.6	0.0500	25.0	1.4	25.69	
182	4.3000	28.8	0.0500	25.0	1.2	28.07	
186	4.3500	30.4	0.0500	25.0	1.6	29.53	
189	4.4000	31.9	0.0500	25.0	1.5	31.55	
192	4.4500	33.4	0.0500	25.0	1.5	34.93	
196	4.5000	35.5	0.0500	25.0	2.1	39.18	
199	4.5500	37.1	0.0500	25.0	1.6	43.07	
203	4.6000	39.9	0.0500	25.0	2.8	47.53	
207	4.6500	42.4	0.0500	25.0	2.5	53.65	
211	4.7000	45.2	0.0500	25.0	2.8	58.69	
214	4.7500	48.4	0.0500	25.0	3.2	54.57	
218	4.8000	52.5	0.0500	25.0	4.1	45.07	
223	4.8500	57.9	0.0500	25.0	5.4	161.38	
226	4.9000	64.2	0.0500	25.0	6.3	330.59	
235	4.9500	75.1	0.0500	25.0	10.9	480.08	
250	5.0000	102.2	0.0500	25.0	27.1	548.69	
	5.013073	117.0				548.91	
256	5.0500	158.7	0.0500	25.0	56.5	516.25	
259	5.1000	172.8	0.0500	25.0	14.1	394.32	
263	5.1500	179.7	0.0500	25.0	6.9	220.82	
272	5.2000	181.8	0.0500	25.0	2.1	62.95	
275	5.2500	184.9	0.0500	25.0	3.1	-4.80	
278	5.3000	187.6	0.0500	25.0	2.7		
282	5.3500	189.4	0.0500	25.0	1.8		
286	5.4000	191.0	0.0500	25.0	1.6		
289	5.4500	192.5	0.0500	25.0	1.5		
292	5.5000	193.7	0.0500	25.0	1.2		

Titration report

LabX

Titration (EQP) [1] Scope 1/1, Sample 5/6

E - V & dE/dV - V Curve Scope 1/1, Sample 5/6



Measured Values Scope 1/1, Sample 5/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-64.1		25.0			
10	2.0000	-3.8	2.0000	25.0	60.3		
43	2.0500	-4.1	0.0500	25.0	-0.3		
46	2.1000	-3.4	0.0500	25.0	0.7		
50	2.1500	-2.7	0.0500	25.0	0.7		
53	2.2000	-2.2	0.0500	25.0	0.5	11.00	
57	2.2500	-1.4	0.0500	25.0	0.8	13.24	
60	2.3000	-1.0	0.0500	25.0	0.4	12.20	
64	2.3500	-0.2	0.0500	25.0	0.8	12.18	
67	2.4000	0.2	0.0500	25.0	0.4	12.30	
70	2.4500	1.0	0.0500	25.0	0.8	12.10	
73	2.5000	1.6	0.0500	25.0	0.6	12.74	
76	2.5500	2.1	0.0500	25.0	0.5	12.27	
79	2.6000	2.8	0.0500	25.0	0.7	12.57	
83	2.6500	3.4	0.0500	25.0	0.6	11.77	
86	2.7000	4.1	0.0500	25.0	0.7	11.56	
89	2.7500	4.5	0.0500	25.0	0.4	11.29	
92	2.8000	5.2	0.0500	25.0	0.7	11.61	
95	2.8500	5.6	0.0500	25.0	0.4	12.08	
98	2.9000	6.3	0.0500	25.0	0.7	12.20	
101	2.9500	7.0	0.0500	25.0	0.7	12.57	
104	3.0000	7.7	0.0500	25.0	0.7	11.83	
108	3.0500	8.1	0.0500	25.0	0.4	12.11	
111	3.1000	8.7	0.0500	25.0	0.6	11.90	
114	3.1500	9.4	0.0500	25.0	0.7	12.23	
117	3.2000	9.9	0.0500	25.0	0.5	13.00	
120	3.2500	10.8	0.0500	25.0	0.9	13.76	
123	3.3000	11.3	0.0500	25.0	0.5	13.64	

Titration report

LabX

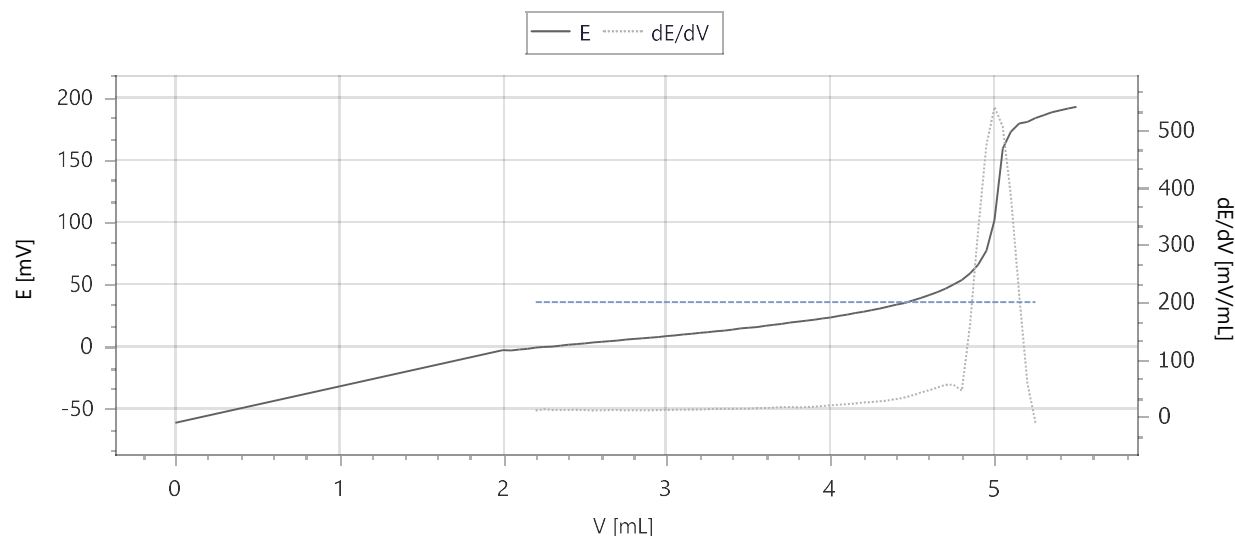
126	3.3500	12.0	0.0500	25.0	0.7	14.24	
129	3.4000	12.8	0.0500	25.0	0.8	14.51	
133	3.4500	13.5	0.0500	25.0	0.7	14.65	
137	3.5000	14.2	0.0500	25.0	0.7	15.47	
140	3.5500	15.1	0.0500	25.0	0.9	15.17	
143	3.6000	15.7	0.0500	25.0	0.6	15.47	
146	3.6500	16.6	0.0500	25.0	0.9	16.05	
149	3.7000	17.4	0.0500	25.0	0.8	15.92	
153	3.7500	18.1	0.0500	25.0	0.7	16.60	
156	3.8000	19.1	0.0500	25.0	1.0	17.33	
160	3.8500	19.9	0.0500	25.0	0.8	17.60	
163	3.9000	20.7	0.0500	25.0	0.8	18.61	
166	3.9500	21.9	0.0500	25.0	1.2	19.95	
170	4.0000	22.7	0.0500	25.0	0.8	20.74	
173	4.0500	23.8	0.0500	25.0	1.1	21.83	
176	4.1000	25.0	0.0500	25.0	1.2	22.19	
179	4.1500	26.2	0.0500	25.0	1.2	23.37	
182	4.2000	27.2	0.0500	25.0	1.0	24.59	
186	4.2500	28.5	0.0500	25.0	1.3	25.23	
190	4.3000	29.8	0.0500	25.0	1.3	26.41	
194	4.3500	31.4	0.0500	25.0	1.6	28.82	
197	4.4000	32.6	0.0500	25.0	1.2	31.86	
200	4.4500	34.3	0.0500	25.0	1.7	33.73	
205	4.5000	36.1	0.0500	25.0	1.8	37.25	
210	4.5500	38.3	0.0500	25.0	2.2	42.29	
212	4.6000	40.4	0.0500	25.0	2.1	47.31	
216	4.6500	42.6	0.0500	25.0	2.2	51.25	
220	4.7000	45.9	0.0500	25.0	3.3	56.63	
226	4.7500	49.2	0.0500	25.0	3.3	53.31	
230	4.8000	52.7	0.0500	25.0	3.5	41.01	
234	4.8500	57.6	0.0500	25.0	4.9	156.81	
237	4.9000	64.6	0.0500	25.0	7.0	332.77	
241	4.9500	75.0	0.0500	25.0	10.4	479.34	
244	5.0000	101.1	0.0500	25.0	26.1	544.61	
	5.013240	116.6				544.81	EQP
251	5.0500	159.7	0.0500	25.0	58.6	509.79	
254	5.1000	173.5	0.0500	25.0	13.8	387.12	
268	5.1500	176.4	0.0500	25.0	2.9	216.44	
271	5.2000	180.7	0.0500	25.0	4.3	60.34	
275	5.2500	184.2	0.0500	25.0	3.5	-4.91	
278	5.3000	186.7	0.0500	25.0	2.5		
281	5.3500	188.7	0.0500	25.0	2.0		
285	5.4000	190.3	0.0500	25.0	1.6		
288	5.4500	191.9	0.0500	25.0	1.6		
291	5.5000	193.2	0.0500	25.0	1.3		

Titration (EQP) [1] Scope 1/1, Sample 6/6

E - V & dE/dV - V Curve Scope 1/1, Sample 6/6

Titration report

LabX



Measured Values Scope 1/1, Sample 6/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-61.2		25.0			
11	2.0000	-2.9	2.0000	25.0	58.3		
44	2.0500	-3.1	0.0500	25.0	-0.2		
48	2.1000	-2.4	0.0500	25.0	0.7		
51	2.1500	-1.8	0.0500	25.0	0.6		
54	2.2000	-1.0	0.0500	25.0	0.8	11.17	
58	2.2500	-0.5	0.0500	25.0	0.5	13.34	
61	2.3000	0.0	0.0500	25.0	0.5	12.07	
64	2.3500	0.6	0.0500	25.0	0.6	12.01	
67	2.4000	1.4	0.0500	25.0	0.8	12.12	
70	2.4500	2.0	0.0500	25.0	0.6	12.29	
73	2.5000	2.4	0.0500	25.0	0.4	11.81	
76	2.5500	3.1	0.0500	25.0	0.7	11.19	
79	2.6000	3.7	0.0500	25.0	0.6	11.47	
82	2.6500	4.2	0.0500	25.0	0.5	11.90	
86	2.7000	4.8	0.0500	25.0	0.6	11.82	
88	2.7500	5.4	0.0500	25.0	0.6	11.28	
92	2.8000	6.1	0.0500	25.0	0.7	11.60	
95	2.8500	6.5	0.0500	25.0	0.4	11.58	
98	2.9000	7.1	0.0500	25.0	0.6	11.60	
101	2.9500	7.7	0.0500	25.0	0.6	11.91	
104	3.0000	8.4	0.0500	25.0	0.7	12.17	
107	3.0500	8.9	0.0500	25.0	0.5	12.40	
110	3.1000	9.6	0.0500	25.0	0.7	12.64	
113	3.1500	10.2	0.0500	25.0	0.6	12.89	
116	3.2000	10.8	0.0500	25.0	0.6	12.85	
119	3.2500	11.5	0.0500	25.0	0.7	13.34	
122	3.3000	12.3	0.0500	25.0	0.8	13.93	
125	3.3500	12.8	0.0500	25.0	0.5	14.06	
128	3.4000	13.5	0.0500	25.0	0.7	14.02	
131	3.4500	14.4	0.0500	25.0	0.9	14.11	

Titration report

LabX

134	3.5000	15.1	0.0500	25.0	0.7	14.43	EQP
137	3.5500	15.6	0.0500	25.0	0.5	15.12	
140	3.6000	16.5	0.0500	25.0	0.9	15.31	
144	3.6500	17.3	0.0500	25.0	0.8	16.07	
147	3.7000	18.1	0.0500	25.0	0.8	16.93	
150	3.7500	19.0	0.0500	25.0	0.9	17.14	
153	3.8000	19.8	0.0500	25.0	0.8	16.53	
156	3.8500	20.7	0.0500	25.0	0.9	16.96	
159	3.9000	21.5	0.0500	25.0	0.8	17.77	
162	3.9500	22.4	0.0500	25.0	0.9	18.97	
165	4.0000	23.3	0.0500	25.0	0.9	20.19	
169	4.0500	24.5	0.0500	25.0	1.2	21.19	
172	4.1000	25.6	0.0500	25.0	1.1	22.28	
175	4.1500	26.7	0.0500	25.0	1.1	23.38	
178	4.2000	27.8	0.0500	25.0	1.1	25.03	
182	4.2500	29.1	0.0500	25.0	1.3	25.94	
185	4.3000	30.5	0.0500	25.0	1.4	27.23	
188	4.3500	32.0	0.0500	25.0	1.5	28.74	
191	4.4000	33.5	0.0500	25.0	1.5	30.88	
194	4.4500	34.8	0.0500	25.0	1.3	33.88	
197	4.5000	36.7	0.0500	25.0	1.9	37.46	
201	4.5500	38.8	0.0500	25.0	2.1	42.47	
205	4.6000	41.1	0.0500	25.0	2.3	46.46	
210	4.6500	43.6	0.0500	25.0	2.5	51.39	
215	4.7000	46.3	0.0500	25.0	2.7	55.73	
221	4.7500	49.7	0.0500	25.0	3.4	56.10	
224	4.8000	53.2	0.0500	25.0	3.5	45.51	
228	4.8500	58.7	0.0500	25.0	5.5	159.24	
234	4.9000	65.5	0.0500	25.0	6.8	325.80	
241	4.9500	76.8	0.0500	25.0	11.3	473.65	
244	5.0000	101.2	0.0500	25.0	24.4	539.96	
	5.013567	116.8				540.11	
251	5.0500	158.6	0.0500	25.0	57.4	505.40	
254	5.1000	172.1	0.0500	25.0	13.5	383.75	
257	5.1500	178.8	0.0500	25.0	6.7	213.58	
268	5.2000	180.0	0.0500	25.0	1.2	58.83	
272	5.2500	183.1	0.0500	25.0	3.1	-10.30	
275	5.3000	185.5	0.0500	25.0	2.4		
278	5.3500	187.6	0.0500	25.0	2.1		
281	5.4000	189.2	0.0500	25.0	1.6		
284	5.4500	190.7	0.0500	25.0	1.5		
287	5.5000	192.0	0.0500	25.0	1.3		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6
Comment	

Titration report

LabX

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
03.02.2026 14:55:41	PredefinedUser	Administrator

(16) Sample suspended